AUTONOMOUS FRAMEWORK FOR REAL-TIME DATA QUALITY, PRIVACY, AND SCHEMA DRIFT IN STREAMING ENVIRONMENTS

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Project Proposal Report

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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning. To the best of our knowledge and belief, it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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ABSTRACT

This research presents an autonomous, integrated framework for real-time data quality management, privacy preservation, and semantic integrity in streaming environments. As modern enterprises increasingly rely on high-velocity data pipelines spanning IoT, healthcare, finance, and social media platforms, ensuring continuous data quality, privacy compliance, and schema consistency has become critically important. Existing solutions address these concerns in isolation — separately handling data cleaning, privacy enforcement, or schema drift — and typically operate reactively on stored data rather than proactively at the point of ingestion.

The proposed framework addresses these limitations through four tightly integrated research components. The first component, led by Silva A.S. (IT22549136), introduces a distributed multi-agent system for autonomous real-time quality assessment and preprocessing of heterogeneous unstructured data streams (text, audio, video, and image), leveraging Apache Kafka for high-throughput ingestion exceeding 100,000 messages per second. The second component, developed by Arachchi L.T.E. (IT22893666), proposes a closed-loop framework for structured streaming data quality that integrates continuous quality measurement, explainable AI-driven preprocessing, quarantine fallback handling, benchmarking, and rollback mechanisms with full audit traceability. The third component, by Weerakoon W.R.W.M.N.S. (IT22204752), introduces an explainable, risk-aware schema drift detection and governance engine for real-time relational and semi-structured data pipelines, incorporating SHAP/LIME interpretability, risk scoring, human-in-the-loop validation, and automated rollback. The fourth component, developed by Arangalla H.A.A.K. (IT22556684), proposes an inline privacy-preserving pipeline that enforces PII detection and anonymization at the data ingestion layer — prior to storage — using differential privacy, homomorphic encryption, and tokenization techniques aligned with GDPR, HIPAA, and CCPA.

Together, these components constitute a unified, compliance-ready, and accountable streaming data governance system. Expected outcomes include measurable improvements in data quality, privacy assurance, schema resilience, and interpretability across enterprise-scale pipelines.

Keywords: Real-time streaming, data quality, multi-agent systems, explainable AI, schema drift, privacy preservation, Apache Kafka, GDPR compliance, human-in-the-loop governance.

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
AWS	Amazon Web Services
CCPA	California Consumer Privacy Act
CDC	Change Data Capture
DP	Differential Privacy
ETL	Extract, Transform, Load
GCP	Google Cloud Platform
GDPR	General Data Protection Regulation
HE	Homomorphic Encryption
HIPAA	Health Insurance Portability and Accountability Act
HITL	Human-in-the-Loop
IoT	Internet of Things
JSON	JavaScript Object Notation
LIME	Local Interpretable Model-Agnostic Explanations
LLM	Large Language Model
MAS	Multi-Agent System
MPC	Multi-Party Computation
NLP	Natural Language Processing
NER	Named Entity Recognition
PII	Personally Identifiable Information
PSNR	Peak Signal-to-Noise Ratio
RF	Random Forest
SHAP	SHapley Additive exPlanations
SLIIT	Sri Lanka Institute of Information Technology
SSIM	Structural Similarity Index Measure

XAI	Explainable Artificial Intelligence
XAI	Explainable Artificial Intelligence

1. INTRODUCTION

1.1 Background

Modern enterprises depend increasingly on real-time data pipelines to power analytics, decision-making, and operational systems. With the exponential growth of high-velocity data streams from diverse sources—including IoT sensors, social media platforms, healthcare systems, financial networks, and enterprise applications—ensuring the quality, privacy, and structural integrity of these streams has become both critical and challenging [1].

This group research addresses four tightly interrelated problems in real-time streaming data management, each forming a distinct research component:

Component 1: Autonomous Multi-Agent Framework for Real-Time Quality Assessment and Preprocessing of Unstructured Data Streams (Silva A.S., IT22549136)

The emergence of multi-agent systems (MASs) in artificial intelligence has transformed approaches to real-time data quality management, particularly for unstructured data streams generated by modern enterprise systems, social media, IoT devices, and multimedia platforms. Traditional centralized systems are increasingly insufficient to handle the scale, heterogeneity, and velocity of these data flows. Recent advances in Large Language Models (LLMs) have enhanced MAS capabilities by enabling agents to perform semantic understanding, contextual reasoning, and inter-agent dialogue [2]. These LLM-driven agents can dynamically adjust roles, coordinate tasks, and engage in iterative evaluation loops to maintain high data quality in complex environments. Event-driven orchestration frameworks, such as Apache Kafka, enable high-throughput ingestion, real-time routing, and fault-tolerant processing across text, audio, video, and image modalities [3].

Component 2: Real-Time Data Quality Assessment and Explainable Preprocessing Framework with Benchmarking and Rollback Support (Arachchi L.T.E., IT22893666)

Organizations rely on real-time data pipelines to power analytics and operational systems. Conventional data quality solutions operating in offline or batch-processing contexts are insufficient for dynamic environments where even minor data errors can cascade into significant downstream effects [4]. The integration of Explainable Artificial Intelligence (XAI) has emerged as a promising approach, making data transformations interpretable and thereby strengthening accountability and user trust [5]. Incorporating rollback mechanisms and audit trails further ensures regulatory compliance and operational safety in industries such as healthcare, finance, and IoT-driven systems.

Component 3: Explainable Schema Drift Detection and Governance in Real-Time Data Pipelines (Weerakoon W.R.W.M.N.S., IT22204752)

Research in real-time data management and schema evolution has accelerated with the growing use of high-volume, heterogeneous data pipelines across IoT, finance, and healthcare domains. Schema drift—which occurs when an evolving dataset has unanticipated changes in structure such as adding, removing, renaming, or changing the type of a column—causes significant barriers to consistency, governance, and system reliability [6]. Standard solutions utilizing static schema validation rules or offline reconciliations are insufficient in high-throughput streaming contexts where schema changes can propagate almost instantaneously through dependent applications [7]. Explainable AI techniques such as SHAP and LIME have been examined as means of increasing interpretability of anomaly detection and drift management [8].

Component 4: Real-Time Privacy Preservation in Data Pipelines for Streaming Environments (Arangalla H.A.A.K., IT22556684)

The exponential growth of real-time data from digital ecosystems—IoT, healthcare, financial platforms, and social media—amplifies risks to privacy, particularly given the increasing variety and velocity of structured and unstructured data streams [9]. Preserving data privacy across diverse modalities in high-throughput environments has become essential for both ethical compliance and regulatory adherence under frameworks such as GDPR, HIPAA, and CCPA [10]. A wide range of cryptographic, statistical, and governance-based techniques have been proposed, including differential privacy (DP), homomorphic encryption (HE), secure multi-party computation (MPC), federated learning (FL), and synthetic data generation. Despite their promise, most existing approaches apply privacy preservation mechanisms only after data has been ingested and stored, leaving raw sensitive information at rest vulnerable to insider threats and breaches [11].

1.2 Research Scope

This research collectively addresses four complementary problems in real-time streaming data governance:

- Component 1 implements a real-time, distributed multi-agent pipeline for autonomous data quality assessment and preprocessing at the ingestion layer, specifically optimized for heterogeneous unstructured data streams including text, audio, video, and image modalities. The framework employs a modular event-driven architecture with a novel dual-validation mechanism enforcing both initial and post-processing integrity checks at scale.
- Component 2 addresses structured streaming data quality by integrating continuous quality measurement, explainable preprocessing, quarantine fallback handling, benchmarking, and rollback capabilities into a single closed-loop framework for high-velocity structured data pipelines.
- Component 3 focuses on schema drift in relational and semi-structured data streams, developing a risk-aware, interpretable detection and governance engine that incorporates automated rollback, human-in-the-loop validation, and XAI-based explanations of drift events.
- Component 4 develops an inline, privacy-first pipeline that enforces PII detection and anonymization at the point of data ingestion—before storage—using a combination of statistical, cryptographic, and semantic privacy techniques, ensuring that raw sensitive information never persists in enterprise storage systems.

1.3 Literature Review

1.3.1 Introduction

Contemporary data engineering faces simultaneous challenges across data quality, structural consistency, and privacy. Literature in each sub-domain has evolved considerably, yet integrated solutions that address all three dimensions in a real-time streaming context remain scarce. The following subsections synthesize key findings across each research component.

1.3.2 Multi-Agent Systems and Unstructured Data Quality

Distributed MASs have evolved from isolated, single-agent models into collaborative ecosystems capable of autonomous decision-making, negotiation, and adaptive workflow optimization [2]. Recent advances in LLMs have enhanced MAS capabilities, enabling agents to perform semantic understanding, contextual reasoning, and inter-agent dialogue. Modern MAS architectures support centralized, decentralized, and hierarchical configurations, allowing scalable deployment across heterogeneous infrastructures. Key research emphasizes integrating automated governance, dual-stage validation, and mechanism design to address challenges inherent in distributed pipelines. Multi-agent reinforcement learning, ensemble collaboration, and incentive-compatible frameworks have been proposed to optimize cooperation, fairness, and reliability [3].

1.3.3 Structured Data Quality in Streaming Environments

Real-time data quality management for structured streams has seen significant methodological development. Rule-based systems provide formal consistency guarantees but require extensive manual specification and lack adaptability across evolving domains [12]. Statistical and ML-based repair approaches, including matrix factorization imputation, deep generative methods, and HoloClean's probabilistic graphical models, achieve strong correction accuracy but share limitations for streaming deployment due to computational complexity [13]. The importance of inter-column dependencies for correction quality has been established in the data cleaning literature, with dependency-aware architectures proving more transparent and efficient than implicit constraint-based formulations [14]. Human-in-the-Loop (HITL) approaches have been applied in data integration and entity resolution, with research establishing that human review is necessary when automated systems face ambiguous cases [15]. Explainable AI techniques—particularly SHAP and LIME—have been integrated to provide interpretable explanations for automated decisions, improving auditor and practitioner trust in automated preprocessing systems.

1.3.4 Schema Drift Detection and Management

Schema drift denotes unintended anomalous changes to database schema such as attribute addition, deletion, renaming, or type alteration occurring without systemic versioning or data modeling practices [6]. Significant progress has been made in schema drift research within structured environments using hybrid detection approaches, statistical profiling, and knowledge-based annotations. State-of-the-art systems including schema induction models and automated drift incident platforms have demonstrated success in detecting entity-level and semantic changes. However, such solutions are often constrained by their use of annotated datasets, pre-defined schemas, and assumed well-structured input data, inhibiting adaptation to real-time, dynamic, heterogeneous data [7]. Most current tools focus on logging and alerts or automated schema adjustments without providing proactive risk estimation, interpretable recommendations, or user-guided governance actions. A major gap exists in frameworks that simultaneously detect schema drift in real-time, assign interpretable risk-informed scores to drift events, and support selective human-in-the-loop feedback with rollback mechanisms [8].

1.3.5 Privacy Preservation in Data Pipelines

A wide range of cryptographic, statistical, and governance-based privacy techniques have been proposed for data pipelines. Differential privacy (DP) provides formal mathematical guarantees of privacy by adding calibrated noise to data, protecting individual records while preserving aggregate statistical properties [10]. Homomorphic encryption (HE) enables computation on encrypted data without decryption, though at substantial computational cost. Federated learning (FL) allows model training across distributed data sources without centralizing raw data.

Tokenization and role/attribute-based access controls provide governance-level safeguards. Despite their promise, most existing approaches apply privacy mechanisms only after data has been ingested and stored, leaving a critical vulnerability window during transmission and initial ingestion [11]. Named Entity Recognition (NER) has been applied for automated PII detection in unstructured text, while structural analysis handles PII in tabular and semi-structured formats. Existing pipeline tools such as Protecto AI, Airflow, Dagster, DataGrail, and OneTrust provide post-ingestion privacy capabilities but do not enforce privacy at the point of data entry.

1.3.6 Synthesis and Research Gaps

Across all four research components, a common pattern emerges: existing solutions are reactive rather than proactive, operating on stored data rather than enforcing quality, privacy, and structural integrity at the ingestion layer. Furthermore, most existing approaches address these concerns in isolation, creating fragmented governance landscapes that fail to provide end-to-end accountability. The proposed research directly addresses these systemic gaps by architecting an integrated, four-component framework that enforces quality, privacy, and schema governance simultaneously at the point of data ingestion.

1.4 Research Gap

The following research gaps motivate this study:

Gap 1: Absence of Autonomous Unstructured Data Quality Frameworks (IT22549136)

- No fully autonomous, scalable, and context-sensitive frameworks exist for heterogeneous unstructured data streams spanning text, audio, video, and image modalities.
- Existing approaches often require human oversight for edge cases and cannot integrate comprehensive dual-stage validation across all data modalities.
- Incentive compatibility and long-term reliability in multi-agent ecosystems are inadequately addressed in large-scale deployments.

Gap 2: Lack of Integrated, Explainable, and Reversible Structured Data Quality Systems (IT22893666)

- Current industry and academic approaches focus on isolated aspects such as anomaly detection or automated cleaning, but rarely provide an integrated, transparent, and auditable solution.
- Existing systems generally lack quarantine fallback mechanisms, version-controlled rollback, and XAI-based explanations for automated preprocessing decisions.

Gap 3: Insufficient Real-Time, Risk-Aware Schema Drift Governance (IT22204752)

- Most schema drift systems operate reactively in batch mode and cannot detect or classify drift events in real-time streaming contexts.
- Existing detection tools lack risk scoring, user-in-the-loop governance, explainable drift analysis, and rollback mechanisms for unsafe schema changes.
- Heterogeneous data sources (relational, JSON, streaming logs) are rarely handled within a single unified drift detection framework.

Gap 4: Absence of Ingestion-Layer Privacy Enforcement (IT22556684)

- Existing privacy preservation mechanisms act after data has been stored, leaving raw PII vulnerable to insider threats, credential theft, and misconfigurations during the ingestion phase.
- No existing framework enforces inline, real-time privacy protection across both structured and unstructured data streams prior to any persistent storage event.

1.5 Research Problem

The overarching research problem addressed by this group project is:

"How can an integrated, autonomous framework be designed and implemented to simultaneously ensure real-time data quality, privacy, and structural integrity in high-velocity streaming environments, across both structured and unstructured data modalities, without sacrificing performance, interpretability, or compliance?"

Each component addresses a specific dimension of this problem:

- Component 1 (IT22549136): How can a distributed multi-agent framework autonomously assess and preprocess unstructured data quality in real time, ensuring scalability, reliability, and end-to-end integrity without human intervention?
- Component 2 (IT22893666): How can an integrated, transparent, and auditable framework simultaneously ensure real-time data quality measurement, explainable preprocessing, quarantine handling, benchmarking, and reversible corrections for structured streaming data?
- Component 3 (IT22204752): How can an explainable, risk-aware schema drift detection and governance system provide real-time, interpretable, and safe schema management across heterogeneous data pipelines?
- Component 4 (IT22556684): How can a pipeline enforce privacy preservation at the ingestion layer to ensure that raw PII never exists at rest in enterprise data environments, while maintaining regulatory compliance and low-latency performance?

1.6 Research Questions

Component 1 (IT22549136):

- What mechanisms enable autonomous, real-time quality assessment across diverse unstructured data sources in distributed multi-agent systems?
- How can event-driven MAS architectures be optimized for scalability, fault tolerance, and compliance in high-throughput environments?
- Which strategies best integrate dual-stage validation and game-theoretic incentive alignment without human intervention?

- How do automated governance and machine learning protocols impact long-term data integrity and reliability?

Component 2 (IT22893666):

- How can real-time data quality be continuously measured across structured streaming pipelines using dimensions of completeness, consistency, accuracy, and uniqueness?
- How can XAI techniques be integrated into automated preprocessing to ensure interpretable, auditable, and reversible data corrections?
- What fallback and benchmarking mechanisms can sustain pipeline continuity and validate the effectiveness of preprocessing interventions?

Component 3 (IT22204752):

- How can real-time structural changes across heterogeneous data sources be effectively detected, including column addition, removal, renaming, and type changes?
- What methodologies can assign risk scores to detected schema changes, allowing high-impact drifts to be prioritized over low-impact or inconsequential alterations?
- How can human-in-the-loop governance be incorporated into a real-time schema drift pipeline, ensuring safe schema evolution without impacting performance?

Component 4 (IT22556684):

- What is the difference in security posture between ingestion-layer privacy enforcement and post-ingestion PII protection in enterprise data environments?
- How does enforcing privacy at ingestion improve resilience against insider threats and regulatory compliance under GDPR, HIPAA, and CCPA?
- What are the implications on latency and throughput when enforcing privacy preservation at the ingestion layer?

1.7 Objectives

1.7.1 Main Objective

To design, implement, and validate an integrated autonomous framework for real-time data quality management, privacy preservation, and semantic integrity in streaming environments, combining distributed multi-agent quality assessment, explainable structured data cleaning, risk-aware schema drift governance, and ingestion-layer privacy enforcement into a unified, compliance-ready pipeline architecture.

1.7.2 Specific Objectives by Component

Component 1 — Autonomous Multi-Agent Unstructured Data Quality (IT22549136):

- Design and implement a high-throughput Apache Kafka-based ingestion layer capable of processing real-time data streams exceeding 100,000 messages per second from social media APIs, IoT sensor networks, document repositories, and multimedia platforms.
- Construct an event-driven orchestration layer implementing the orchestrator-worker pattern with intelligent content-based partitioning, dynamic load balancing, and Byzantine fault-tolerant protocols.
- Develop four specialized AI agents optimized for domain-specific quality assessment: Text Processing Agent (linguistic accuracy, semantic consistency), Audio Processing Agent (signal-to-noise ratio, word error rate), Video Processing Agent (SSIM, PSNR, temporal consistency), and Image Processing Agent (resolution, exposure, distortion detection).
- Implement a dual-validation framework integrating real-time assessment with post-processing validation to achieve comprehensive quality assurance without human intervention.

Component 2 — Explainable Structured Data Quality with Rollback (IT22893666):

- Develop a real-time data quality measurement module continuously monitoring streaming records against completeness, consistency, accuracy, and uniqueness dimensions.
- Implement explainable automated preprocessing using XAI techniques (SHAP, LIME) to provide interpretable justifications for every corrective action applied to streaming data.
- Design a quarantine fallback mechanism that isolates suspicious records for offline review while maintaining pipeline continuity.
- Build benchmarking and impact analysis capabilities to validate quality improvements before and after preprocessing, and implement rollback mechanisms with full audit trail support.

Component 3 — Schema Drift Detection and Governance (IT22204752):

- Curate and annotate a benchmark dataset of schema drift instances covering addition, removal, renaming, and type changes from relational and semi-structured data sources.
- Develop agile drift detection algorithms capable of generalizing to previously unseen schemas and dynamic pipeline environments.
- Design and validate a risk scoring tool to classify schema drifts by severity impact, applying XAI techniques for interpretable insights.
- Develop a human-in-the-loop governance process for high-risk drift events and implement robust rollback and recovery mechanisms for safe schema state reversion.

Component 4 — Ingestion-Layer Privacy Preservation (IT22556684):

- Design an inline privacy-preserving pipeline that enforces PII detection and anonymization at the point of data ingestion prior to any persistent storage event.
- Incorporate layered privacy techniques—semantic PII detection, NER, differential privacy, homomorphic encryption, and tokenization—tailored to structured, semi-structured, and unstructured data streams.

- Ensure regulatory compliance with GDPR, HIPAA, and CCPA while maintaining configurable privacy rules and low-latency performance in high-throughput environments.
- Evaluate the framework's security posture improvements versus existing post-ingestion privacy tools and measure latency and throughput implications.

2. METHODOLOGY

2.1 Overall System Architecture

The integrated framework adopts a layered microservices architecture that decomposes the streaming data governance pipeline into four independently scalable, yet tightly coordinated components. At the infrastructure level, data from external sources—structured databases, CSV/APIs, unstructured documents, and real-time Kafka/IoT streams—flows through a Data Ingestion and Buffering Layer supporting both batch ingestion (Apache Airflow) and real-time ingestion (Apache Kafka). The central orchestrator then routes data to the appropriate specialized engine: the Structured AI Data Engine, the Unstructured AI Agents Engine, the Data Privacy and Anonymization Engine, and the Language and Semantic Engine. Processed data is stored in a Data Warehouse (BigQuery, Snowflake) and Data Lake (S3, ADLS, Delta Lake). A unified User Dashboard and API layer provides Data Quality Scorecards, Explainable AI Reports, Privacy Audit Trails, Profile and Schema Explorers, and RESTful/GraphQL endpoints for downstream consumers including BI tools, ML models, and enterprise applications.

2.2 Component Methodologies

2.2.1 Component 1: Multi-Agent Framework for Unstructured Data Quality (IT22549136)

The methodology for Component 1 implements a five-layer autonomous pipeline:

- Layer 1 — Data Ingestion: An Apache Kafka-based ingestion layer processes heterogeneous unstructured streams from social media APIs, IoT networks, document repositories, and multimedia platforms at rates exceeding 100,000 messages per second. Adaptive ingestion pipelines maintain 99.2% uptime under schema evolution.
- Layer 2 — Event-Driven Orchestration: An orchestrator-worker pattern with content-based partitioning and dynamic load balancing routes incoming data to modality-specific agents. The Orchestrina agent provides Byzantine fault-

tolerant protocols maintaining system integrity even with up to one-third compromised agents.

- Layer 3 — Autonomous Processing: Four specialized AI agents perform modality-specific quality assessment: Text (NLP accuracy, semantic consistency), Audio (signal-to-noise ratio, word error rate <10%), Video (SSIM, PSNR, temporal consistency), and Image (resolution, exposure, distortion detection).
- Layer 4 — Dual-Validation: A two-stage validation protocol applies quality checks at both initial ingestion and post-processing, ensuring only validated data is published downstream.
- Layer 5 — Output Integration: Validated streams are published to dedicated Kafka topics for downstream business analytics, ML pipelines, and enterprise applications.

Evaluation metrics include agent accuracy targets for each modality, multi-agent coordination latency, task completion rate, and load balancing variance. Data sources include public datasets (Kaggle, ImageNet, COCO), synthetic GAN/VAE-generated multimodal streams, and real-time APIs for benchmarking.

2.2.2 Component 2: Structured Data Quality Framework (IT22893666)

The Component 2 methodology implements a five-interconnected-component closed-loop architecture:

- Real-Time Data Quality Measurement: Continuously monitors streaming records against completeness, consistency, accuracy, and uniqueness using a composite quality scoring mechanism (Completeness 40%, Validity 40%, Consistency 20%) normalized to a 0–95 scale with grade mapping.
- Explainable Automated Preprocessing: Applies corrective methods (imputation, normalization, outlier correction) while generating SHAP and LIME explanations for every fix decision, ensuring each correction is interpretable and traceable to either a named reviewer or a timed auto-apply fallback.

- User-Unavailable Handling (Quarantine Table Fallback): Suspicious records are isolated in a quarantine table pending user review, ensuring pipeline continuity without data loss.
- Benchmarking and Impact Analysis: Re-assesses data quality before and after preprocessing and detects unintended side effects, with before-and-after quality delta feedback for reviewers.
- Rollback and Traceability: Implements version control and full audit trails enabling reversibility, compliance, and transparency for every preprocessing decision.

LLM-based validation rule extraction (LLaMA 3.3-70B via Groq) automates constraint generation from unstructured domain documentation, providing domain-adaptive rules without manual specification. The Human-in-the-Loop governance layer uses a React/Vite frontend with inactivity detection (pointer, keyboard, mouse, and touch event tracking) to trigger automated fallback with mandatory approval confirmation.

2.2.3 Component 3: Schema Drift Detection and Governance (IT22204752)

The Component 3 methodology follows a systematic multi-stage pipeline:

- Data Collection and Preparation: Heterogeneous datasets from relational (SQL) and semi-structured (JSON, streaming logs) sources are assembled. Schema evolution scenarios covering column addition, deletion, renaming, and type changes are simulated with annotations for training and evaluation.
- Schema Extraction and Profiling: Schemas are automatically extracted and versioned over time to support longitudinal comparison. Semi-structured data is flattened to a consistent internal representation.
- Drift Detection: Automated schema comparison algorithms identify structural drift including column/field additions or deletions, attribute renames, data type changes, and structural changes in semi-structured data.

- Risk Scoring: Detected drifts are classified by operational impact severity using interpretable risk scoring models, distinguishing high-impact events from inconsequential alterations.
- Human-in-the-Loop Governance: High-risk drift events are isolated for expert validation. A governance interface allows reviewers to approve, reject, or modify proposed schema changes.
- XAI Explanations: SHAP and LIME are applied to provide interpretable explanations of drift causes, impacts, and recommended actions for data engineers and governance teams.
- Rollback and Recovery: Safe schema state reversion mechanisms maintain transaction consistency and schema version history, enabling recovery from erroneous or risky drift events.

2.2.4 Component 4: Ingestion-Layer Privacy Preservation (IT22556684)

The Component 4 methodology implements inline privacy enforcement through the following pipeline stages:

- Data Collection and Ingestion: Multi-source data ingestion using Apache Kafka handles structured (databases, CSV, APIs), semi-structured (JSON, XML), and unstructured (text, images, audio) streams in real time.
- PII Detection: A multi-modal PII detection engine combining Named Entity Recognition (NER) for text, structural analysis for tabular data, and semantic classifiers for semi-structured content identifies sensitive information prior to any storage event.
- Privacy Enforcement Layer: Layered privacy techniques are applied inline: Differential Privacy (DP) for aggregate protection, Homomorphic Encryption (HE) for computation on encrypted data, Tokenization for identifier replacement, and Synthetic Data Generation for high-sensitivity cases.
- Compliance Validation: Configurable rule sets aligned with GDPR, HIPAA, and CCPA govern privacy enforcement decisions, with audit logs capturing all detection and anonymization events.

- Protected Data Storage: Only privacy-protected data is persisted in downstream data stores, ensuring raw PII never exists at rest.
- Performance Evaluation: Latency and throughput benchmarks compare the framework against existing tools (Protecto AI, Airflow, Dagster, DataGrail, OneTrust) to demonstrate feasibility for high-throughput streaming environments.

2.3 Gantt Chart

The project is planned across 12 months (August 2025 – July 2026) with the following major milestones:

Phase	Activity	Duration
Phase 1	Literature Review & Proposal Finalization	Aug – Sep 2025
Phase 2	System Architecture Design & Framework Setup	Oct – Nov 2025
Phase 3	Component Development (Parallel)	Dec 2025 – Mar 2026
Phase 4	Integration & System Testing	Apr – May 2026
Phase 5	Evaluation, Benchmarking & Documentation	Jun – Jul 2026

3. PROJECT REQUIREMENTS

3.1 Functional Requirements

The integrated framework must fulfill the following functional requirements across all four components:

Data Ingestion and Streaming

- The system shall ingest heterogeneous data streams—including social media feeds, IoT sensor networks, document repositories, multimedia content, relational databases, and JSON/XML sources—in real time using Apache Kafka topics.
- The architecture shall support ingestion rates exceeding 100,000 messages per second with adaptive load balancing and fault tolerance.

Autonomous Quality Assessment (Component 1)

- The system shall autonomously assess unstructured data quality through four specialized modal agents (text, audio, video, image) using domain-specific metrics.
- A dual-validation protocol shall enforce both initial ingestion and post-processing quality checks, fully automated and agent-driven.

Structured Data Quality Management (Component 2)

- The system shall continuously measure data quality across completeness, consistency, accuracy, and uniqueness dimensions, generating composite quality scores on a normalized scale.
- Automated preprocessing corrections shall be accompanied by SHAP/LIME explanations. Suspicious records shall be quarantined pending user review, with rollback available for any preprocessing action.

Schema Drift Management (Component 3)

- The system shall detect structural schema changes (addition, deletion, renaming, type changes) in real time across relational and semi-structured sources.
- Each detected drift shall be assigned a risk score, with high-risk events routed to a human governance interface. Rollback mechanisms shall enable safe schema state reversion.

Privacy Preservation (Component 4)

- The system shall enforce PII detection and anonymization inline at the ingestion layer, ensuring raw PII is never persisted in any storage system.
- The privacy engine shall support differential privacy, homomorphic encryption, tokenization, and NER-based detection, configurable per regulatory framework (GDPR, HIPAA, CCPA).

3.2 Non-Functional Requirements

Requirement	Specification
Performance	Ingestion throughput >100,000 messages/sec; preprocessing latency <50ms per record
Scalability	Horizontal scaling across Kafka partitions; auto-scaling in cloud environments
Reliability	≥99.2% system uptime; Byzantine fault-tolerant agent coordination
Security	End-to-end encryption; role-based access control; full audit logging
Interpretability	SHAP/LIME explanations for every automated correction and drift decision
Compliance	GDPR, HIPAA, CCPA alignment; configurable regulatory rule sets
Maintainability	Modular microservices architecture; versioned schema and rollback support

3.3 Proposed Budget

Item	Description	Estimated Cost (LKR)
Cloud Computing	AWS/GCP/Azure for Kafka, storage, ML training, and benchmarking	120,000
Software Licenses	Development tools, APIs, and monitoring platforms	15,000
LLM API Access	Groq/LLaMA API credits for rule extraction and agent intelligence	25,000
Data Acquisition	Public dataset access and synthetic data generation tools	10,000
Documentation & Printing	Proposal and final report printing and binding	8,000
Miscellaneous	Contingency and sundry expenses	12,000
TOTAL		190,000

4. DESCRIPTION OF PERSONNEL AND FACILITIES

The research will be conducted by four undergraduate students specializing in Data Science at the Sri Lanka Institute of Information Technology (SLIIT), under the supervision of Dr. Lakmini Abeywardhana and Dr. Mahima Weerasinghe of the Faculty of Computing.

Student	Student ID	Research Component
Silva A.S. (Amodhya Sharinda Silva)	IT22549136	Autonomous Multi-Agent Framework for Unstructured Data Quality
Arachchi L.T.E.	IT22893666	Real-Time Structured Data Quality with XAI and Rollback
Weerakoon W.R.W.M.N.S.	IT22204752	Schema Drift Detection and Governance
Arangalla H.A.A.K.	IT22556684	Ingestion-Layer Privacy Preservation

Each team member possesses expertise in data engineering, real-time data processing, machine learning, and their respective research specialization. The project benefits from continuous academic guidance and technical support from both supervisors throughout the research lifecycle.

The research environment comprises both personal computing resources and scalable institutional facilities. Individual resources support initial data stream processing, experimentation, and prototype development. Cloud infrastructure (AWS/GCP/Azure) enables large-scale streaming data monitoring, real-time processing, and performance evaluation. Institutional facilities provide workflow management tools, schema version tracking platforms, and visualization interfaces for the complete research workflow.

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